| INCH-POUND |

MIL-T-27/306A
13 March 1991
SUPERSEDING
MIL-T-27/306
1 DECEMBER 1980

MILITARY SPECIFICATION SHEET

# TRANSFORMER, SATURABLE, TF5S40ZZ

Ⓐ | Inactive for new design after the date of this specification. |

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation. MIL-T-27.

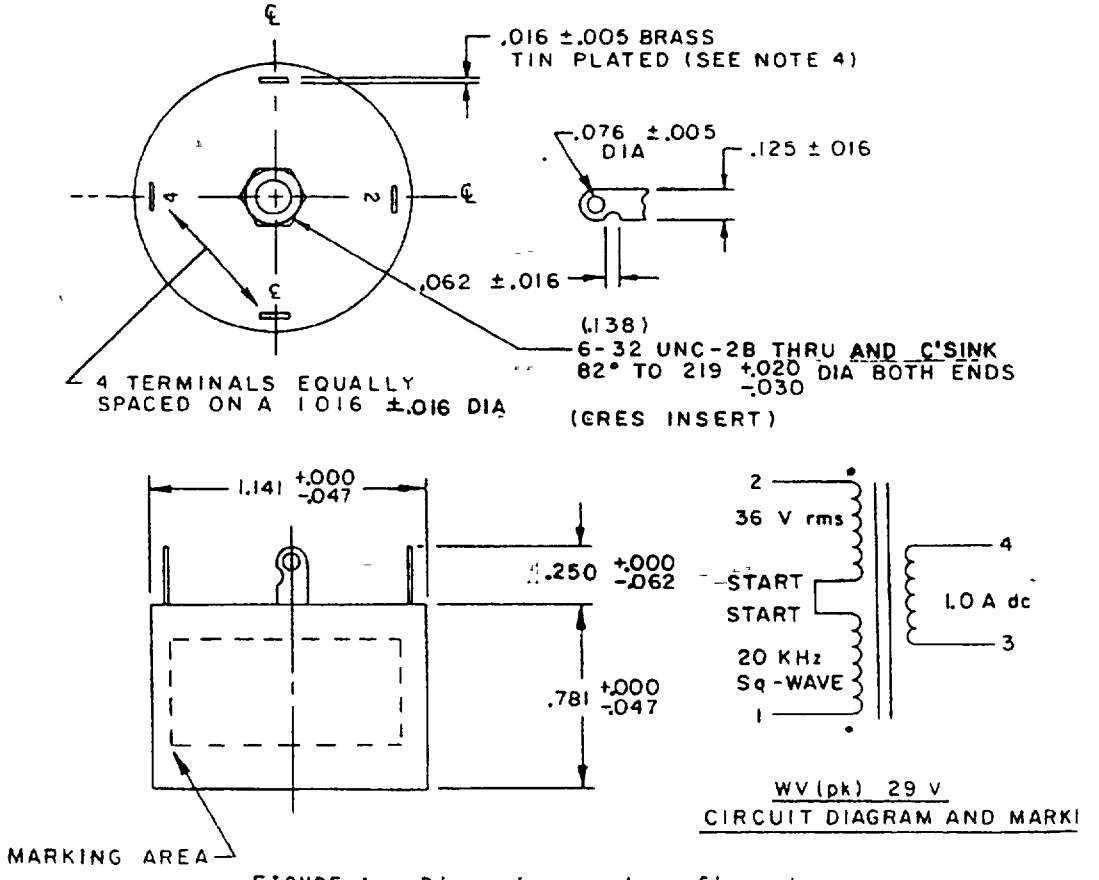



FIGURE 1. Dimensions and configurations.

Ⓐ denotes changes


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| Inches | mm | Inches | mm |
|---|---|---|---|
| .005 | 0.13 | .125 | 3.18 |
| .016 | 0.41 | .138 | 3.61 |
| .020 | 0.51 | .219 | 5.56 |
| .030 | 0.76 | .250 | 6.35 |
| .047 | 1.19 | .781 | 19.84 |
| 062 | 1.57 | 1.016 | 25.81 |
| .076 | 1.93 | 1.141 | 28.98 |

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the side of the case.
4. Brass, 70 percent Cu, 30 percent Zn in accordance with Federal Specification QQ-B-613.

FIGURE 1. Dimensions and configurations - Continued.



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REQUIREMENTS: (When numbers in parentheses, i e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Voltage.

(1-GND): 36 volts rms (input), 20 kilohertz square wave.
(2-GND) 17 to 20 volts.

Current.

(1-2): 10.5 milliamperes.
(3-4): 1 0 amperes dc.

Volt-ampere rating: 0.304 volt-ampere

Primary impedance (1-2): 3,100 ohms.

Working voltage (peak). 29 volts.

Design and construction:

Dimensions and configuration. See figure 1.

Duty cycle: Continuous

Case: Encapsulated.

Material: Epoxy.

Terminals: Solder lug.

Terminal height: 0.250 inch, maximum.

Weight. 36.86 grams, maximum.

Operating temperature range: -55°C to +130°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 2.0 pounds.

Dielectric withstanding voltage (at sea level) 100 volts rms.

Electrical characteristics

Rated load (see figure 2) With 36 volts rms, 20 kilohertz square wave across (1-GND), current in (1-2) shall not exceed 10.5 milliamperes dc.

Voltage across (2-GND): 17 to 20 volts.

Current across (3-4): 1.0 ampere dc.

DC resistance:

(1-2). 12.8 to 17.2 ohms.
(3-4) 0.0025 to 0 0035 ohm.

Temperature rise 50°C with 36 volts rms, 20 kilohertz square wave across (1-GND) and with 17 to 20 volts across (2-GND) at an ambient temperature of 80°C.



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Vibration (high frequency): MIL-STD-202, method 204, test condition A.

Shock (specified pulse): MIL-STD-202, method 213, test condition H.

Ⓐ Quality assurance provisions:

Qualification inspection. Not applicable.

Quality conformance inspection: Groups A and B tests of MIL-T-27 shall be applicable.

Marking location: See figure 1.

Part number: M27/306-01.

## CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - AR, MI
Navy - OS, SH
Air Force - 17, 99
DLA - ES

User activities
Army - ME
Navy - AS, MC
Air Force - 19

Preparing activity
Army - ER

Agent
DLA - ES

(Project 5950-0753-50)



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